

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CE/IT)

November 2013

CE216 Computer Organization & Peripherals

Date: 11.11.2013, Monday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) What is the difference between a direct and an indirect address instruction? How many references to memory are needed for each type of instruction to bring an operand into processor register? [04]

(b) Show the block diagram of the hardware that implements the following register transfer statement: [03]

$yT_2: R2 \leftarrow R1, R1 \leftarrow R2$

Q - 2 (a) Draw and explain 4-bit combinational circuit shifter. [06]

(b) What is assembly language? What are the basic rules of the language? [04]

(c) Explain the difference between hardwired control and micro-programmed control. Is it possible to have a hardwired control associated with a control memory? [04]

OR

Q - 2 (a) What the purpose of microinstruction? Explain 16-bit microinstruction format with an example. [06]

(b) What are the basic computer programming categories? Explain any one of them in detail. [04]

(c) Draw and explain Accumulator logic design. [04]

Q - 3 Answer the following (Any Four) [14]

- (1) Draw and explain 4-bit adder-subtractor circuit
- (2) Explain control unit of basic computer.
- (3) Show the complete logic of the interrupt flip-flops R in the basic computer. Use a JK flip-flop and minimize the number of gates.
- (4) Explain assembler process and also list assembler function.
- (5) List assembly language program generated by a compiler from the following program.

```
SUM = 0
SUM = SUM + A + B
DIF = DIF - C
SUM = SUM + DIF
```

SECTION – II

- Q - 4 (a)** What is interrupt? What are the basic types of Interrupts and explain any one of them in brief. [04]
- (b)** A nonpipeline system takes 50 ns to process a task. The same task can be processed in a six-segment pipeline with a clock cycle of 10 ns. Determine the speedup ratio of the pipeline for 100 tasks. What is the maximum speedup that can be achieved? [03]
- Q - 5 (a)** Answer the following (Any Four) [14]
- (1) List and explain Basic computer instruction formats.
 - (2) Find Quotient Q and Remainder A for value 373 divided by 15. Assume 5-bit registers that hold signed numbers.
 - (3) Give Flynn's Classification. Describe one of them with an example.
 - (4) Show how the instruction cycle in the CPU can be processed with a four-segment pipeline.
 - (5) Show the step-by-step multiplication process using Booth algorithm when $(7) * (-5)$ numbers are multiplied. Assume 4-bit registers that hold signed numbers.
- Q - 6 (a)** Answer the following (Any Three) [12]
- (1) Design a 4-bit combinational circuit decremter using four full-adder circuits.
 - (2) Draw and explain the flowchart for interrupt cycle.
 - (3) Explain through the block diagram show that how control memory and the associated hardware needed for selecting the next microinstruction address.
 - (4) Covert the following arithmetic expressions from reverse Polish notation to infix notation.
 - a. $ABCDE * / - +$
 - b. $ABC * / D - EF / +$
 - c. $ABCDE + * - /$
 - d. $ABCDEF + * + * + *$
- (b)** Derive floating point representation of $(-13.9)_{10}$ using 8-bit word. [02]
